CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (IT)

May 2013

IT701 Distributed Computing & Systems

Date: 07.05.2013, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. *All questions are compulsory.*
2. *Indicate clearly, the option you attempt along with its respective question number.*
3. *Figures to right indicate marks.*
4. *Rough work is done in the last page of main answer sheet; don't write anything on the question paper.*

SECTION – I

Q - 1 (a) What is InetAddress? List and explain the factory methods of InetAddress. [05]

(b) Explain getSendBufferSize() and setSendBufferSize() methods of Socket. [02]

Q - 2 (a) What is ServerSocket? Explain the methods of ServerSocket and List the exceptions may be generated by using those methods. [04]

(b) Write a TCP java program that implements a simple client/server application. The client sends is the string (includes alphabets and numbers) to a server. The server receives the data, find the count of alphabets and numbers and send counts separately to the client. [07]

(c) Why in java there are two separate classes like DatagramSocket and DatagramPacket for UDP? [03]

OR

Q - 2 (a) List and explain the methods of Socket class related to IP address and port no. [04]

(b) Write a UDP java program in which client sends integer value and server converts it into binary and returns result to client. [07]

(c) What is MultiCastSocket? Explain joinGroup method of it. [03]

Q - 3 (a) Write a "RandomNumberService" RMI application. The RandomNumber interface should have one remote method called fetchRandomNumber that should return a random number. Write appropriate implementation class, server class and client class. [06]

OR

Q - 3 (a) Write a "FactorialService" RMI application. The Factorial interface should have one remote method called calculateFactorial that should accept one int parameter and should return factorial of the received number parameter. Write appropriate implementation class, server class and client class. [06]

(b) What is marshaling and unmarshaling in RMI? What is the use of rmic and rmiregistry? [04]

(c) Explain object serialization? What is remote activation? [04]

SECTION – II

Q - 4	(a)	What is web service? How does it provide a way to make platform independent service?	[04]
	(b)	What is XML? How XML is used in web services?	[03]
Q - 5	(a)	What is distributed computing? What are the challenges in developing distributed computing applications?	[03]
	(b)	Explain blocking/non-blocking and synchronous/asynchronous primitives in distributed computing.	[04]
	(c)	What is vector clock? List and explain the basic property of vector clock.	[07]
		OR	
	(c)	Write a web service which returns day time to the client.	[07]
Q - 6	(a)	What is consistent global state, strongly consistent, inconsistent state a model of distributed executions?	[06]
		OR	
Q - 6	(a)	What are the algorithmic challenges in distributed computing?	[06]
	(b)	Explain Causal precedence relation. Describe Logical vs. physical concurrency.	[04]
	(c)	Why is it difficult to keep a synchronized system of physical clocks in distributed systems?	[04]